Lab 6, Act 1: Reading the Surface

How much do fumigation trials tell us about a lethal dose?

NRES 746

Answer key. Answers are in magenta.

The scenario

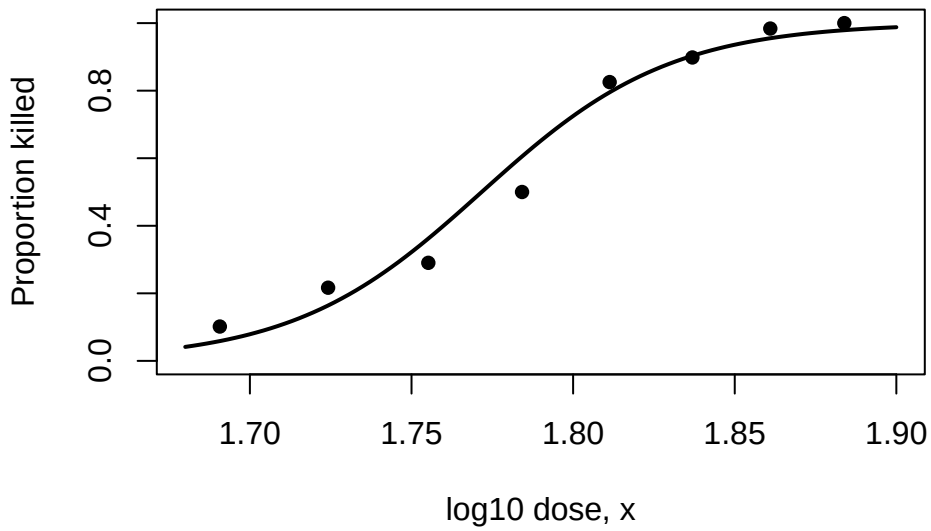
Flour beetles (*Tribolium confusum*) infest stored grain, and fumigants are a standard control. In a classic toxicology experiment, batches of about 60 beetles were exposed to one of eight concentrations of carbon disulphide gas, and the number killed in each batch was recorded (Bliss 1935). Dose-response experiments like this one are used across agriculture, environmental chemistry, pharmacology, and ecotoxicology. Here are the observed measurements:

log10 dose, x	Dose (mg/L)	Beetles exposed	Beetles killed
1.6907	49.1	59	6
1.7242	53.0	60	13
1.7552	56.9	62	18
1.7842	60.8	56	28
1.8113	64.8	63	52
1.8369	68.7	59	53
1.8610	72.6	62	61
1.8839	76.5	60	60

A standard model is logistic in log dose. The probability that a beetle exposed to log dose x dies is

$$p(x) = \frac{1}{1 + e^{-(a+bx)}}, \qquad \text{equivalently} \qquad \log \frac{p(x)}{1 - p(x)} = a + bx,$$

and the number killed at each dose is binomial with that probability. The MLEs are $\hat{a} = -60.7$ and $\hat{b} = 34.3$:

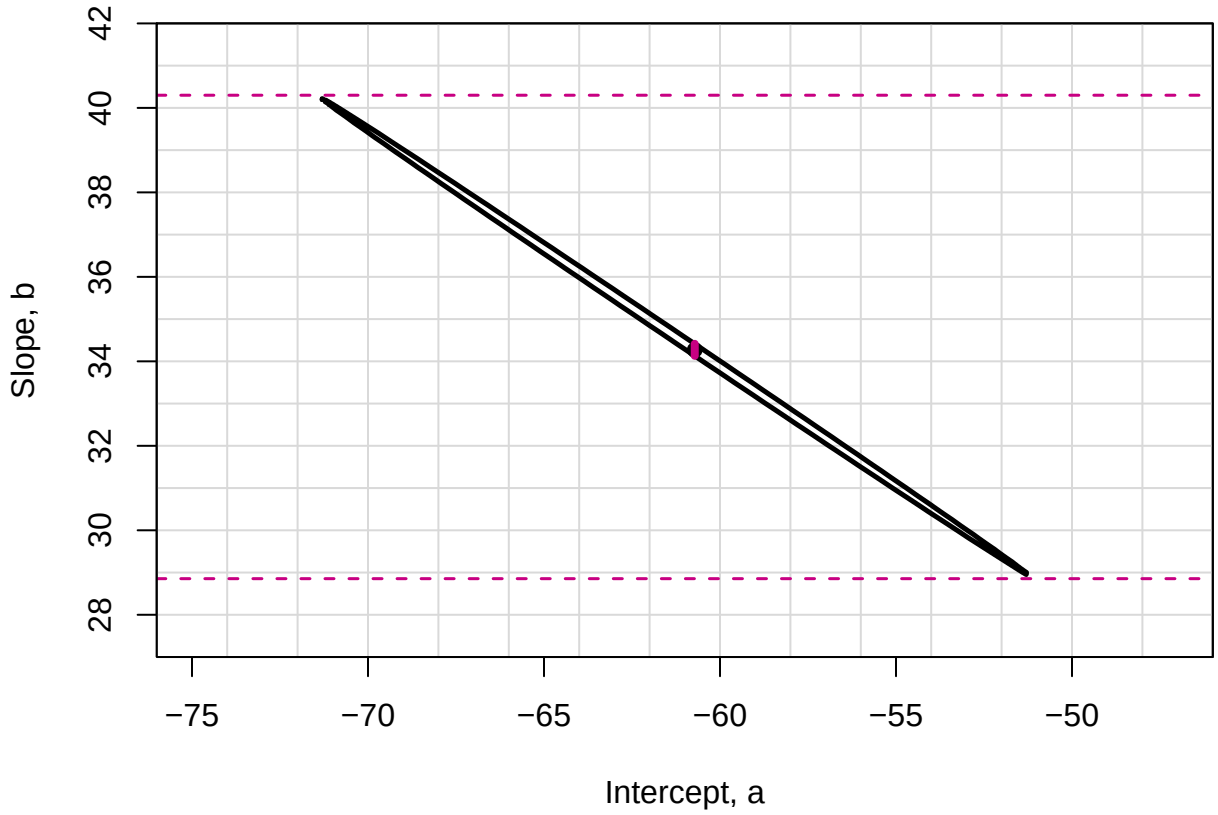


What dose kills half the beetles, and how uncertain is that estimate? This Act works on the second half of that question, using the shape of the likelihood.

Please keep laptops closed for this Act (a calculator or phone is fine).

Part 1: The likelihood surface (individual, ~20 min)

The figure shows the negative log-likelihood (NLL) over (a, b), minus its minimum. The thick curve connects parameter pairs whose NLL is 1.92 above the minimum (the likelihood-ratio cutoff from Monday's lecture). The dot is the MLE.



1a. Estimate two intervals for b by eye from the figure (rough values are fine). The **slice** interval: hold a at

$\hat{a}$ and find the range of b that stays inside the thick curve. The **profile** interval: the full range of b values the thick curve reaches. Which interval is honest, and why?

Slice: 34.13 to 34.41 (essentially the single point 34.3) Profile: 28.9 to 40.3 (roughly 28-29 to 40-41 is fine; both marked in magenta)

The profile. The slice treats a as known; the profile lets a adjust to each b, which is what the uncertainty in b really is when a is also unknown.

1b. The intercept a is the log-odds of death at what dose (as x, and in mg/L)? Use your answer to explain why the valley is a long, thin diagonal: why does every good fit with a steeper slope also need a more negative intercept?

a is the log-odds of death at $x = 0$, i.e. 1 mg/L, far below the tested doses (49 to 77 mg/L). Every good fit must pass near 50% mortality in the middle of the data ($x \approx 1.77$), so a steeper line pivoting around that point must drop further by $x = 0$: larger b needs a more negative a.

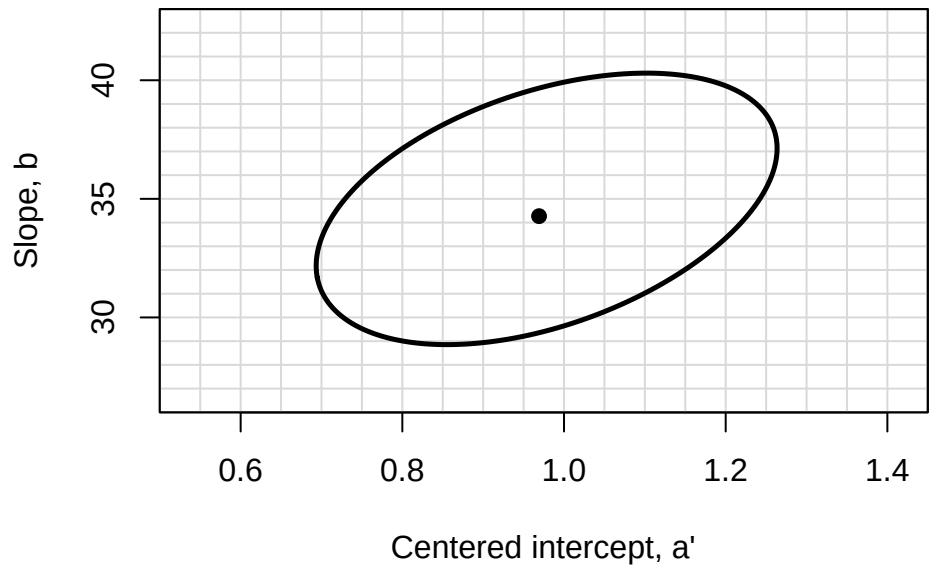
Part 2: Curvature with two parameters (individual, ~20 min)

With one parameter, SE $= 1/\sqrt{\text{curvature}}$. With two parameters, `optim(..., hessian = TRUE)` returns the **Hessian**, a 2×2 table of second derivatives of the NLL at the MLE:

$$H = \begin{pmatrix} H_{aa} & H_{ab} \\ H_{ab} & H_{bb} \end{pmatrix}.$$

The diagonal entries are the curvature along each parameter's axis. The off-diagonal entry H_{ab} measures how the slope in one direction changes as you move in the other direction (the tilt of the valley).

Now **center** the dose: let $x' = x - 1.8$ and fit $\log \frac{p}{1-p} = a' + bx'$. This describes the same set of curves (with $a' = a + 1.8\,b$), so the fitted curve and $\hat{b}$ don't change. Here is the new surface:



For the centered fit, `optim()` reports (rows and columns in the order a', b):

$$H = \begin{pmatrix} 58.5 & -1.26 \\ -1.26 & 0.145 \end{pmatrix}.$$

Here are two candidate standard errors for $\hat{b}$:

$$\text{SE}_1 = \frac{1}{\sqrt{H_{bb}}} \qquad \text{SE}_2 = \frac{1}{\sqrt{H_{bb}}} \times \frac{1}{\sqrt{1 - r^2}}, \quad \text{where } r = \frac{-H_{ab}}{\sqrt{H_{aa} H_{bb}}}.$$

SE_1 uses only b's own curvature, H_{bb}, and ignores the off-diagonal entry. SE_2 uses the off-diagonal entry H_{ab} through r.

2a. Compute SE_1, r, and SE_2, and the approximate 95% interval $\hat{b} \pm 1.96\,\mathrm{SE}$ for each (use $\hat{b} = 34.27$). Which do you think better characterizes our uncertainty about b, and why? Relate your answer to Part 1.

$\mathrm{SE}_1 = 1/\sqrt{0.145} = 2.63$; $r = 1.26/\sqrt{58.5 \times 0.145} = 0.433$; $\mathrm{SE}_2 = 2.63/\sqrt{1 - 0.433^2} = 2.91$

$\mathrm{SE}_1 = 2.63$ $r = 0.433$ $\mathrm{SE}_2 = 2.91$

Interval using SE_1: 29.1 to 39.4 Using SE_2: 28.6 to 40.0

SE_2: its interval is close to the profile interval from 1a (28.9 to 40.3). SE_1 ignores the trade-off between a' and b, like the slice.

2b. For the original, uncentered fit, $H_{bb} = 185$ and $r = -0.99968$. Compute both standard errors and their intervals again. Compare them with your slice and profile intervals from 1a, and with your answers to 2a. Centering didn't change the fitted curve, so what did it change?

$\mathrm{SE}_1 = 1/\sqrt{185} = 0.0735$; $\mathrm{SE}_2 = 0.0735/\sqrt{1 - 0.99968^2} = 2.91$

$\mathrm{SE}_1 = 0.0735$ $\mathrm{SE}_2 = 2.91$

Interval using SE_1: 34.13 to 34.41 Using SE_2: 28.6 to 40.0

SE_1 now matches the (far too narrow) slice; SE_2 matches the profile and is essentially the same as in 2a. Centering changed the shape and tilt of the surface (correlation -0.9997 before, 0.43 after) and the meaning of the intercept (a' = log-odds of death at $x = 1.8$, about 63 mg/L). It didn't change the fitted curve, $\hat{b}$, or the real uncertainty in b.

Group discussion and revision (~10 min)

Compare your answers with your group, especially your explanation in 1b and your answer to 2a, and revise as needed. Record any changes you made, new insights, and any "muddiest points" or topics you want to review.

Open-ended.